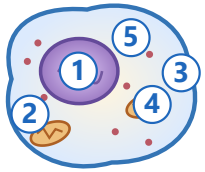
Cells

Eukaryotic cells

1 Animal cells



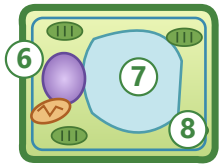
Eukaryotic: DNA enclosed in a nucleus.

1. Nucleus: genes control activities. **2. Cytoplasm:** enzyme-controlled reactions.

3. Membrane: controls entry/exit. **4. Mitochondria:** aerobic respiration transfers energy.

5. Ribosomes: protein synthesis.

2 Plant cells



Have all the animal-cell structures, plus:

6. Cellulose wall: strength/support; also in algae. **7. Permanent vacuole:** cell sap; water keeps the cell firm.

8. Chloroplasts: chlorophyll absorbs light for photosynthesis. Only in photosynthetic cells; normally absent from root hairs.

Compare & specialise

3 Bacterial cells

Prokaryotic: much smaller; no nucleus. Main DNA loop in cytoplasm; some have **plasmids** (extra genes).

Have membrane, cytoplasm, ribosomes and a **non-cellulose wall.** No mitochondria/chloroplasts.

4 Sperm cells



Tail → movement

Tail → swimming. **Mitochondria** → energy for movement.

Acrosome enzymes → penetrate the egg's outer layers.

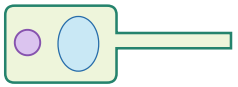
5 Nerve & muscle cells

Nerve: long fibre carries impulses; branched ends connect cells.

Muscle: contractile proteins shorten the cell; mitochondria supply energy for contraction.

Plant adaptations

6 Root hair cells



Extension → large surface area

Long extension → **large absorption area.** Water enters by **osmosis.**

Minerals enter by **active transport** when less concentrated in soil; mitochondria supply energy. Normally no chloroplasts underground.

7 Xylem & phloem

Xylem: dead, hollow tubes carry water/minerals; lignin strengthens walls against collapse.

Phloem: sieve tubes carry dissolved sugars; companion cells supply energy for loading.

Explain **feature** → **effect** → **function,** including in unfamiliar cells.

8 Differentiation

Developing specialised structures for a job. Animals: mainly **early in development;** plants: many cells **throughout life.**

Mature animal cell division mainly repairs/replaces cells. Differentiation is the process; specialisation is the result.